Week 5: VPC Creation and Ec2 Instance Connection

◆ Step 1: Create the VPC

1. Go to **AWS Management Console → VPC**
2. Click **Create VPC**
3. Choose **VPC only**
4. Enter:
   - **Name**: `Custom-VPC`
   - **IPv4 CIDR block**: e.g. `10.0.0.0/16`
5. Click **Create VPC**

◆ Step 2: Create Subnets (Public & Private)

Create subnets in same **Availability Zones** for high availability.

Example:

- **Public Subnet**: `10.0.1.0/24` (AZ-A)
- **Private Subnet**: `10.0.2.0/24` (AZ-A)

Steps:

1. Go to **Subnets → Create subnet**
2. Select your VPC
3. Choose AZ
4. Enter CIDR
5. Create subnet

◆ Step 3: Create an Internet Gateway (IGW)

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name it `Custom-IGW`
4. **Attach** it to your VPC

◆ Step 4: Create Route Tables

You need **separate route tables** for public and private subnets.

Public Route Table

1. Go to **Route Tables → Create route table**
2. Select VPC
3. Add route:
   - Destination: `0.0.0.0/0`
   - Target: **Internet Gateway**
4. Associate with **public subnet**

Private Route Table

- Keep default local route only (no IGW)
- Associate with **private subnet**

◆ Step 5: Enable Auto-Assign Public IP (Public Subnet)

1. Select **Public Subnet**
2. Go to **Edit subnet settings**
3. Enable **Auto-assign public IPv4 address**

◆ Step 6: Configure Security Groups

1. Create a **Security Group**
2. Add inbound rules:
   - HTTP (80) / HTTPS (443)
   - SSH (22) from trusted IP
3. Attach to EC2 instances

◆ Step 7: Launch EC2 Instances

- Public EC2 → Public subnet
- Private EC2 → Private subnet
  Attach correct security groups.

◆ Step 8: Test the Setup

- Public EC2 → Internet access ✓
- Private EC2 → Internet via NAT ✓
- Private EC2 → No direct inbound access ✓

Scenario Based Questions:

1. You have a web application where users access a website, but the database should not be exposed to the internet. How would you design the VPC? Which resources go into public and private subnets?
2. An EC2 instance in a public subnet cannot access the internet. What VPC components would you check and why?
3. Your application server needs to connect to an RDS database securely. How would you configure security groups and subnets?
4. You are asked to design a VPC for 500 servers today, but it should scale to 2,000 servers in the future. How would you choose the CIDR block?
5. A company wants secure connectivity between its on-premises data center and AWS VPC.
   Which AWS services would you choose and why (VPN vs Direct Connect)?
6. You need to connect two VPCs, but both use the same CIDR range. How would you solve this problem?
7. Multiple microservices running in different subnets need to communicate securely. How would you design routing and security groups?
8. Design a VPC for a 3-tier application (Web, App, DB) with high security and scalability. Explain subnets, route tables, gateways, and security groups.
9. You suspect unusual traffic inside your VPC. How would you monitor and analyze network traffic?
10. Why can't a private subnet have an Internet Gateway directly attached?
11. What happens if route tables are misconfigured in a VPC?
12. What happens if a route table has no local route?
13. An EC2 instance allows traffic on port 80 in the security group, but traffic is still blocked.
    What could be the reason?
14. You need to create a VPC that will host 1,000+ EC2 instances across multiple AZs. How do you decide the CIDR block?
15. Only a company's corporate IP should be able to SSH into EC2 instances. How would you implement this securely in AWS VPC?
16. An EC2 instance can send traffic out but cannot receive responses. Which VPC component might be misconfigured and why?